

The 45% Problem — Phase 8 CV Battery Report

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Seed: 20260422 | Folds: 5 | Calibration: 2010-06-11 to 2021-12-31
Hold-out: 2022-01-01 to 2022-12-31

Table 1: Cross-Validation Scores (Calibration Block)

Model	Mean LL ↓	SE (LL)	Mean Brier ↓	SE (Brier)	Mean RPS ↓	SE (RPS)
M0_elo	1.03184	0.02362	0.61933	0.01591	0.21972	0.00349
M1_form	1.06444	0.03293	0.64263	0.02334	0.23014	0.00815
M2_fifa	1.00016	0.01809	0.59788	0.01281	0.20802	0.00618
M3_macro	1.02709	0.02062	0.61597	0.01415	0.21783	0.00426

Table 2: Per-Fold Log-Loss Matrix (5 × 4)

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5
M0_elo	1.02130	1.09190	1.02803	0.95250	1.06548
M1_form	1.05464	1.15227	1.10483	0.95406	1.05637
M2_fifa	0.99911	1.02594	1.03768	0.93345	1.00460
M3_macro	1.01452	1.08292	1.03022	0.95840	1.04938

Table 3: Hold-Out Single-Shot Scores (2022 WC — Never Used for Selection)

Model	Hold-out LL ↓	Hold-out Brier ↓	Hold-out RPS ↓
M0_elo	1.01814	0.60614	0.21488
M1_form	1.10123	0.65921	0.23959
M2_fifa	0.98766	0.58308	0.20398
M3_macro	0.99744	0.59174	0.20724

Table 4: Decision Rule Application (§1.3.5)

Parameter	Value
Primary criterion	mean_cv_log_loss
Ranking (1st→4th)	M2_fifa > M3_macro > M0_elo > M1_form
Winner (M★)	M2_fifa
Winner mean CV LL	1.00016
Runner-up	M3_macro
Runner-up mean CV LL	1.02709
LL gap (abs)	0.02693

LL gap (SE units)	1.489 SE
M★ vs M0 gate (SE)	1.752 SE (threshold ≥ 2.0)
Sanity gate	FAILED
Tie-break invoked	No

Per §1.3.5, M★ = M2_fifa was selected because its calibration log-loss was 1.49 SE below the runner-up (M3_macro). WARNING: sanity gate NOT passed — M★ only 1.75 SE better than M0 (threshold ≥ 2.0 SE).